Name: ________________________ **Date:** ________________________

Collaborators: ________________________ **Lab Section:** ____________

Postlab Exercises (20 points)

Submit the postlab to the TA at the end of the lab.

Newton's First Law (2 points)

Question 3 **1 points**

(a) (1/2 point) What values do you expect for the fractions of momentum and kinetic energy lost,

$$\frac{\Delta p}{p} = ? \qquad \frac{\Delta K}{K} = ?$$

when friction is infinitely large? Explain.

(b) (1/2 point) What values do you expect when there is no friction? Explain.

Question 4 **1 point**

Fill in Data Table 1, where t_i is the transit time through the first photogate and t_f is the transit time through the second.

Data table 1: *Measurements of fractional momentum and energy loss due to friction.*

Trial	t_i	t_f	p_f/p_i	K_f/K_i
1				
2				

For a cart of length ℓ, the velocity at each photogate is $v_i = \ell/t_i$ and $v_f = \ell/t_f$, so the momentum and energy ratios simplify to

$$\frac{p_f}{p_i} = \frac{mv_f}{mv_i} = \frac{t_i}{t_f}, \qquad \frac{K_f}{K_i} = \frac{1/2\, mv_f^2}{1/2\, mv_i^2} = \frac{t_i^2}{t_f^2}.$$

Compute the average ratios and calculate the fractional losses using eqs. (20) and (21). Show your work.

$$\overline{\frac{\Delta p}{p}} = \qquad \overline{\frac{\Delta K}{K}} =$$

Elastic Collisions of Equal and Unequal Masses (6 points)

Question 5 **1 point**

(a) ($1/2$ point) For a perfectly elastic collision, what fractional losses do you expect, given the friction you measured?

$$\frac{\overline{\Delta p}}{p} = \qquad\qquad \frac{\overline{\Delta K}}{K} =$$

(b) ($1/2$ point) Do you expect these values to be the same as or different from those in question 4? Explain.

Question 6 **1 point**

(a) ($1/2$ point) If the two carts have equal length and mass, do the simplified ratio equations from question 4 still apply? Explain.

(b) ($1/2$ point) Based on your answer, which equation will you use to compute $\overline{\Delta p}/p$ and $\overline{\Delta K}/K$ for the elastic collision?

Question 7 **1 point**

Fill in Data Table 2. The launched cart has mass m_1 and the stationary cart has mass m_2. $t_{1,i}$ is the time m_1 takes to pass the first photogate before the collision; $t_{2,f}$ is the time m_2 takes to pass the second photogate after the collision.

Data table 2: *Measurements of elastic collisions between equal masses.*

Trial	m_1	m_2	$t_{1,i}$	$t_{1,f}$	$t_{2,i}$	$t_{2,f}$
1				N/A	N/A	

Question 8 **1 point**

Using your answer to question 6 and the data in Data Table 2, find the fractions of momentum and kinetic energy lost.

$\dfrac{\overline{\Delta p}}{p} =$ $\qquad\qquad$ $\dfrac{\overline{\Delta K}}{K} =$

Question 9 **1 point**

Using the key idea from the Newton's First Law section, determine whether the following inequalities are satisfied:

$$\left| \left(\frac{\overline{\Delta p}}{p}\right)_{\text{Question 8}} - \left(\frac{\overline{\Delta p}}{p}\right)_{\text{Question 4}} \right| > 3\left(\frac{\overline{\Delta p}}{p}\right)_{\text{Question 4}}, \tag{22}$$

and

$$\left| \left(\frac{\overline{\Delta K}}{K}\right)_{\text{Question 8}} - \left(\frac{\overline{\Delta K}}{K}\right)_{\text{Question 4}} \right| > 3\left(\frac{\overline{\Delta K}}{K}\right)_{\text{Question 4}}. \tag{23}$$

In other words, is the fractional loss during the collision more than 3 times larger than the loss due to friction alone?

Question 10 **1 point**

With the stationary cart replaced by a heavier metal mass ($m_1 < m_2$), describe what you observe. How does this collision differ from the equal-mass case? Do the simplified ratio equations still apply? Explain.

Inelastic Collisions (3 points)

Question 11 **1 point**

Record your data for three mass combinations in Data Table 3.

Data table 3: *Measurements of inelastic collisions.*

Trial	t_1	t_2
$m_1 < m_2$		
$m_1 = m_2$		
$m_1 > m_2$		

Question 12 **1 point**

(a) (1/2 point) Which mass combination ($m_1 < m_2$; $m_1 = m_2$; $m_1 > m_2$) produced the smallest change in speed after the collision?

(b) (1/2 point) Why does this combination perform best? Recall from eq. (10) that momentum conservation gives

$$m_1 u_1 = (m_1 + m_2)v.$$

Question 13 **1 point**

What is one factor that limits the final velocity v of the combined carts?

Velocity Measurements with a Ballistic Pendulum (4 points)

Question 14 **1 point**

Record your ballistic pendulum measurements in Data Table 4.

Data table 4: *Measurements of the ballistic pendulum.*

$R =$ ____________________

Ball mass $m =$ ____________________

Pendulum mass $M =$ ____________________

Trial	1	2	3	4	5	$\overline{h}$
Angle θ						—
$h = R(1 - \cos\theta)$						

Question 15 **1 point**

Calculate the mean height $\overline{h}$ and its standard deviation Δh. Show your work and include units.

Question 16 **1 point**

Using eq. (14) with $v_b = u$, find the initial velocity v_b of the ball. Show your work and include units.

Question 17 **1 point**

Find the uncertainty Δv_b using propagation of uncertainties:

$$\frac{\Delta v_b}{v_b} = \frac{\Delta h}{2\overline{h}}.$$

Assume the mass measurements have no uncertainty. Show your work.

Velocity Measurements from Projectile Motion (5 points)

Question 18 **1 point**

Record your measurement of H, its estimated uncertainty ΔH, and all range measurements L in Data Table 5.

Data table 5: *Measurements of projectile motion.*

$H =$ ____________________

$\Delta H =$ ____________________

Trial	1	2	3	4	5	Average $\overline{L}$
Distance L						

Question 19 **1 point**

Calculate the mean range $\overline{L}$ and its standard deviation ΔL. Show your work.

Question 20 **1 point**

Using eq. (17), calculate v_b from H and $\overline{L}$. Show your work.

Question 21 **1 point**

Calculate the uncertainty Δv_b using propagation of uncertainties:

$$\frac{\Delta v_b}{v_b} = \sqrt{\left(\frac{\Delta L}{\overline{L}}\right)^2 + \left(\frac{\Delta H}{2H}\right)^2}.$$

Show your work and include units.

Question 22 **1 point**

(a) ($^1/_2$ point) Are the values of v_b and Δv_b from projectile motion consistent with those from the ballistic pendulum (questions 16 and 17)? That is, do the two results agree within their uncertainties?

(b) ($^1/_2$ point) If the results do not agree, what is one possible explanation?